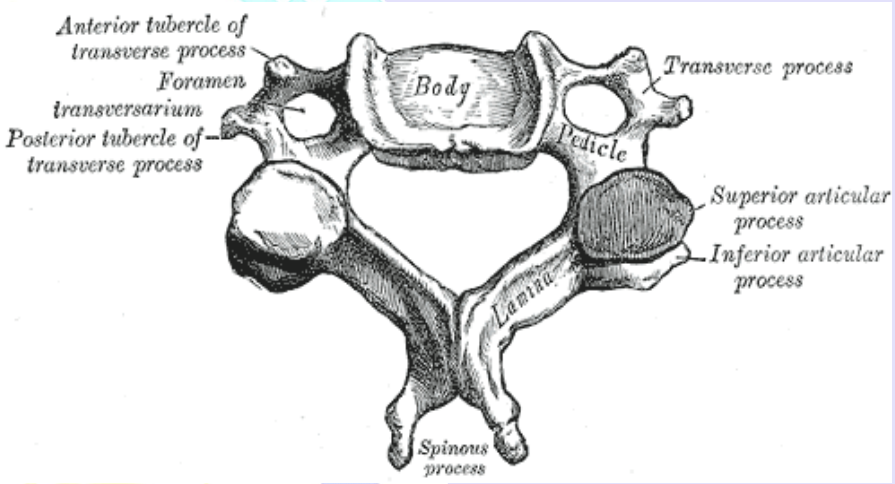
CERVICAL VERTEBRAE

LEARNING OBJECTIVES

- At the end of lecture student should be able to:
  - Identify the cervical spines
  - Differentiate typical and atypical vertebrae
  - Describe bony features of typical & atypical vertebrae
  - Describe important ligament supports
  - Know various movements at different joints
  - Clinical correlation

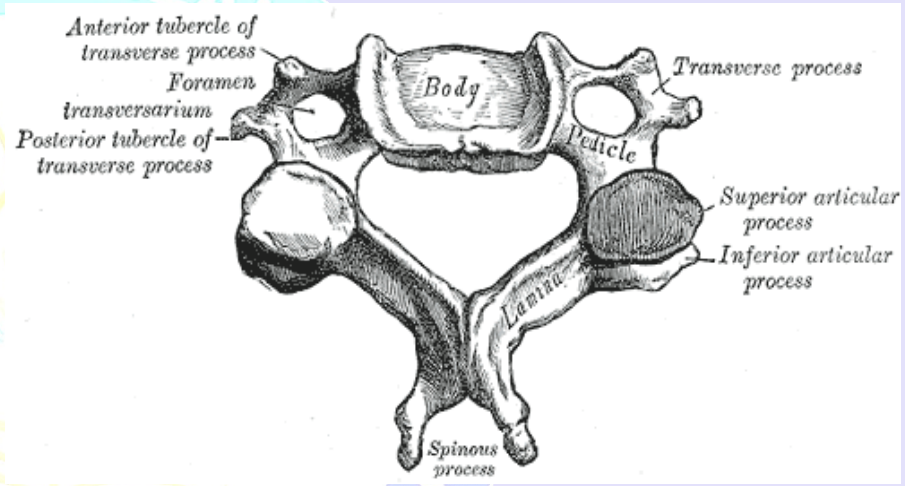
IDENTIFICATION OF CERVICAL SPINES

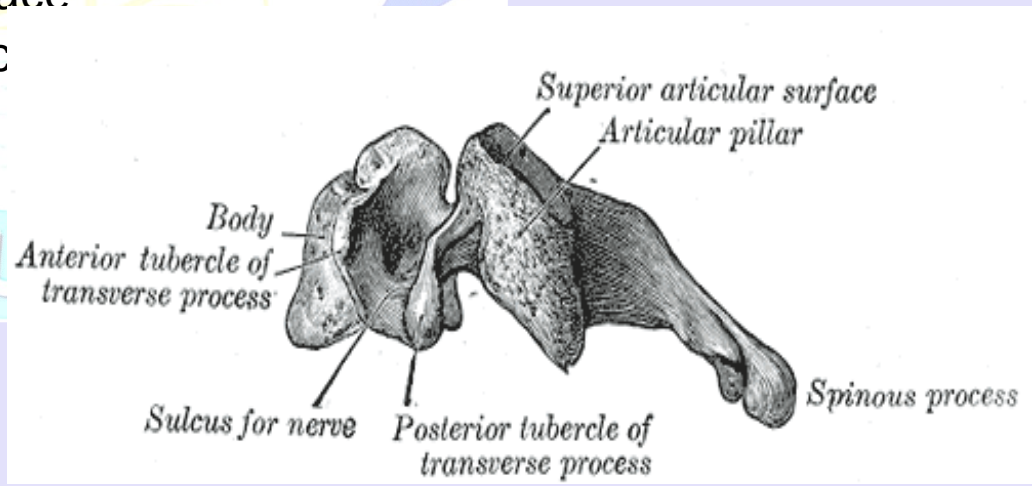
- Total seven cervical spines
- C1 to C7
- **TWO TYPES**
  - Typical (C3---C6)
  - Atypical (C1, C2, C7)
- **FEATURE OF IDENTIFICATION**
  - Foramen transversarium



FEATURES OF TYPICAL VERTEBRAE

- Body
- Vertebral Arch
  - Pedicles
  - Laminae
- Vertebral Foramen
- Spinous Process
- Facets
  - Superior Articulating Surface
  - Inferior Articulating Surfac
- Transverse Process
- Intervertebral Foramen



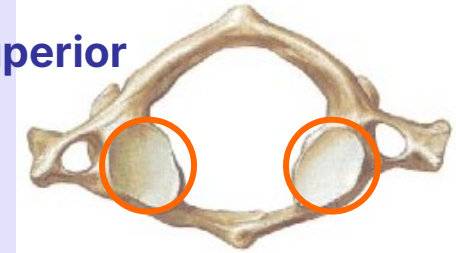


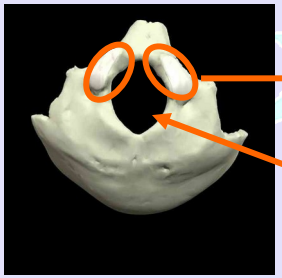
FEATURES OF ATYPICAL VERTEBRAE

Inferior



Superior



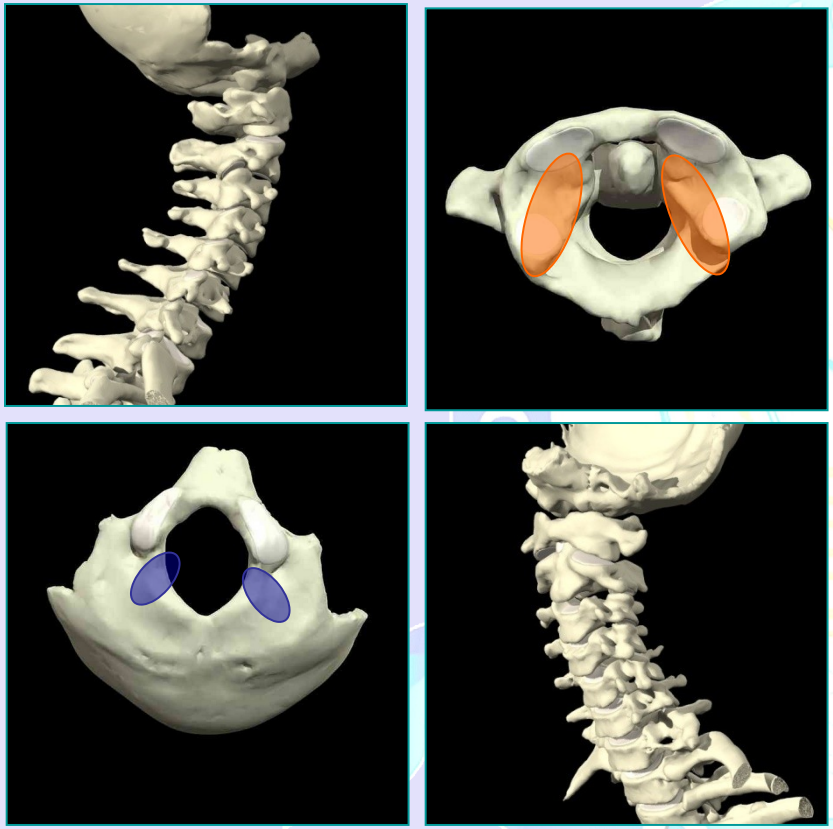


Occipital condyles

Foramen magnum

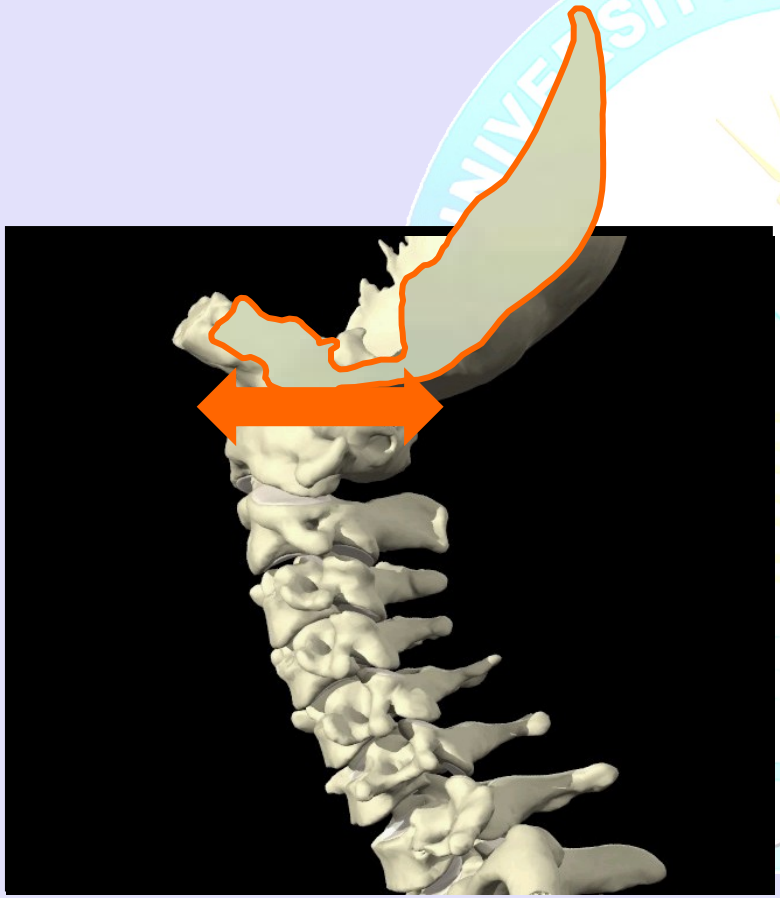
- 1st cervical vertebra
- Aka ***ATLAS***
- Atypical vertebra
- Structure
  - Lack of body & spine
  - Vertebral canal/foramen
  - Anterior arch
  - Transverse process
  - Transverse foramen
  - Anterior tubercle
  - Posterior tubercle
  - Posterior arch
  - Lateral mass

ATLANTO-OCCIPITAL JOINT



- Occiput-C1 segment
  - The occiput-C1 joints are synovial joints comprising the convex occipital condyles, which articulate with the concave lateral masses of C1

MOVEMENT AT ATLANTO-OCCIPITAL JOINT



- Occiput-C1 segment
  - Motion at the occiput-C1 segment is restricted primarily to flexion-extension due to bony structures, ligamentous constraints, and the absence of an intervertebral disc

FEATURES OF ATYPICAL VERTEBRAE

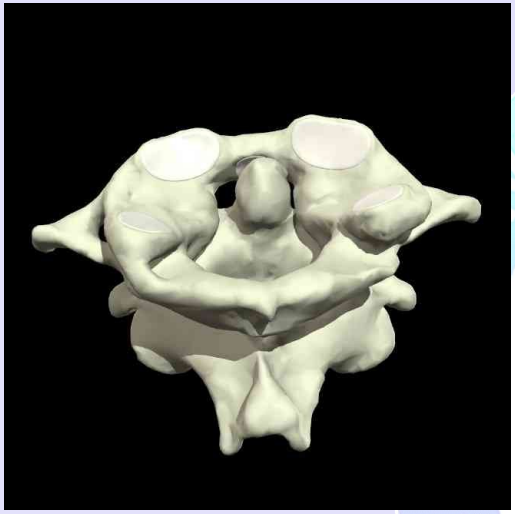
anterior view

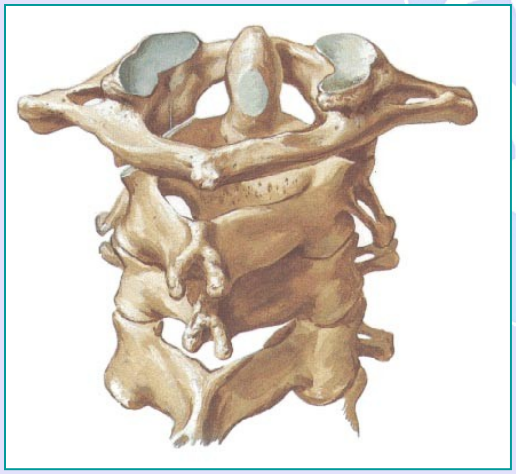
posterior view



- 2nd cervical vertebra
- Aka ***AXIS***
- Odontoid process or dens
- Vertebral canal/foramen
- Facet joints
- Transverse process
- Transverse foramen
- Bifid spinous process
- Lamina

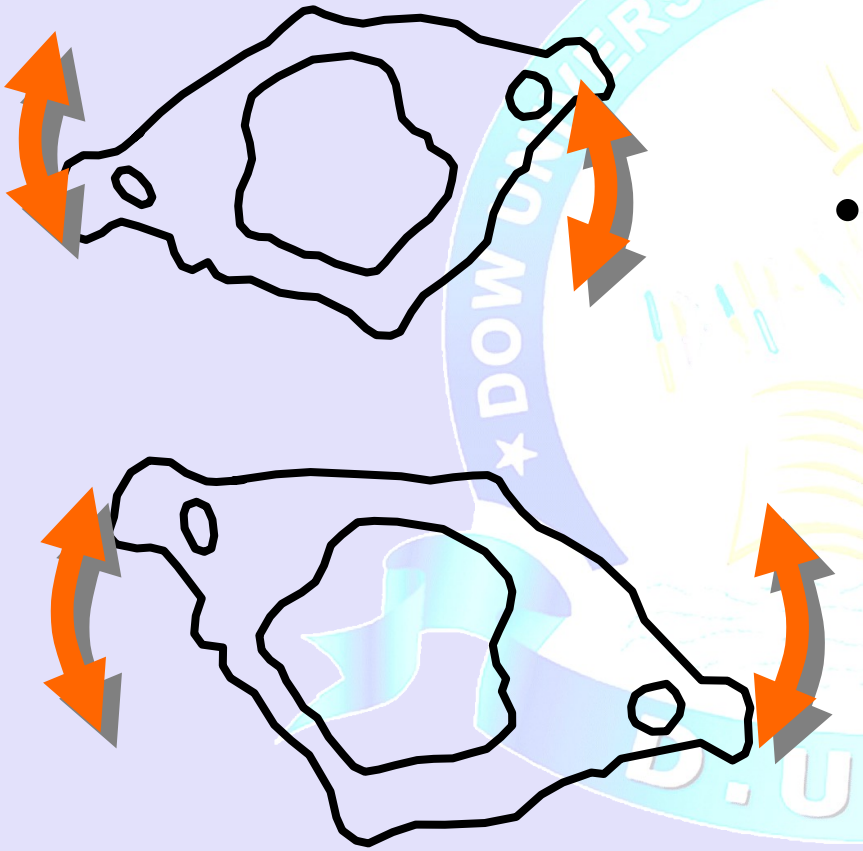
ATLANTO-AXIAL JOINT





- The odontoid process of the axis (C2) articulates with posterior surface of anterior arch of atlas(C1)

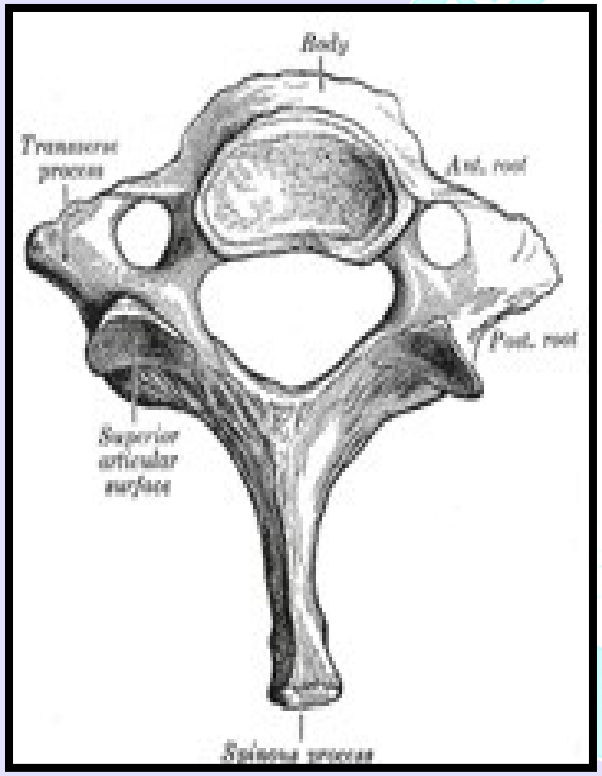
MOVEMENT AT ATLANTO-AXIAL JOINT



- C1-C2 segment
  - The primary motion at the C1-C2 joint is rotation

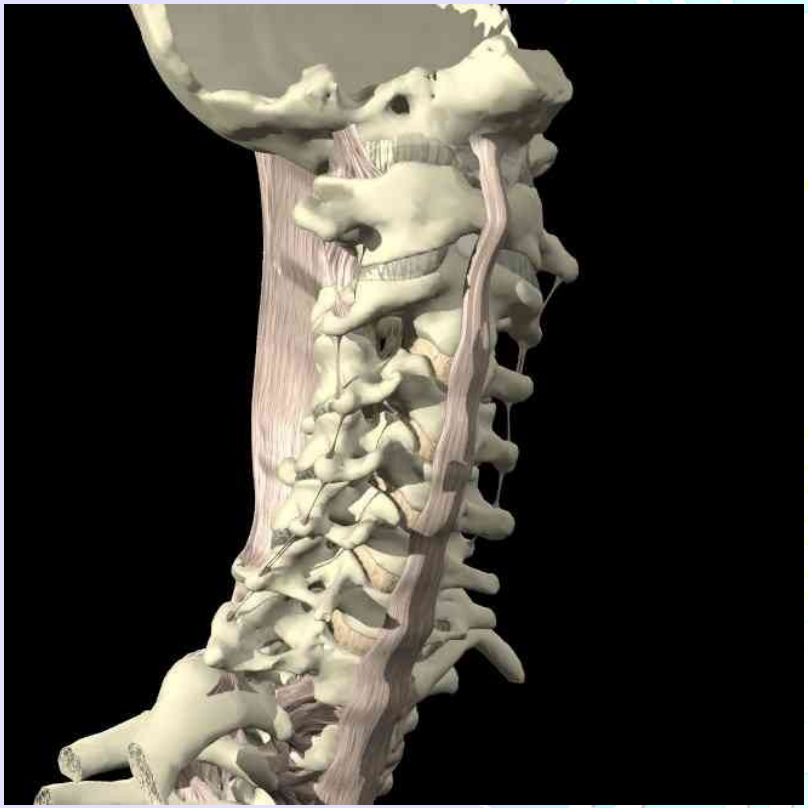


FEATURES OF ATYPICAL VERTEBRAE



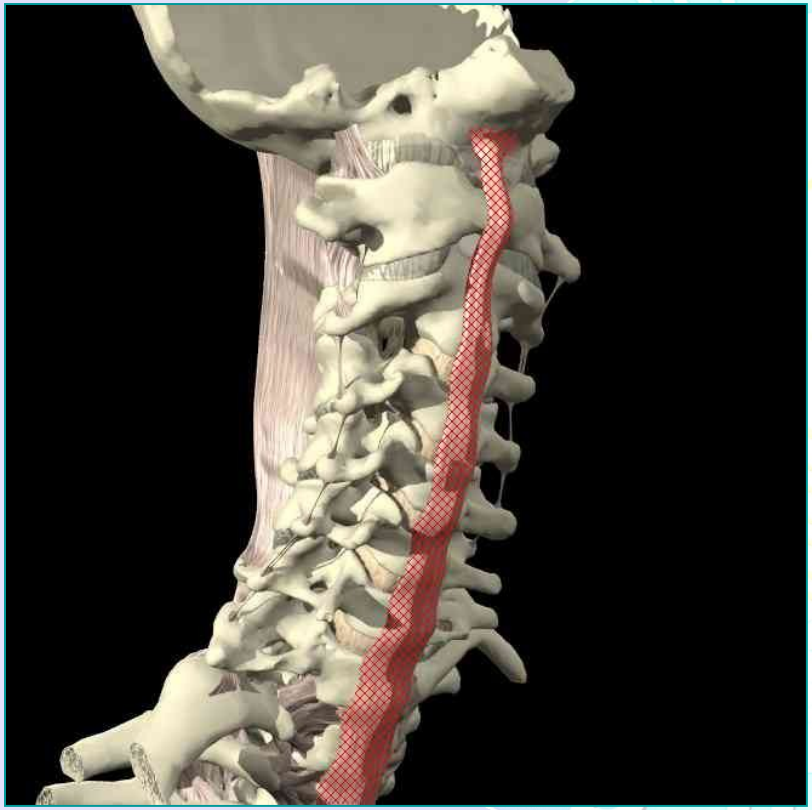
- **The Seventh Cervical Vertebra**
- **vertebra prominens.**
- a long, horizontal and prominent spinous process. Not bifurcated
- The **transverse processes**
  - posterior roots are large and prominent
  - the anterior are small
  - shallow sulcus for the eighth spinal nerve
- The foramen transversarium may be smaller occasionally it is double.

ARRANGEMENT OF LIGAMENTS



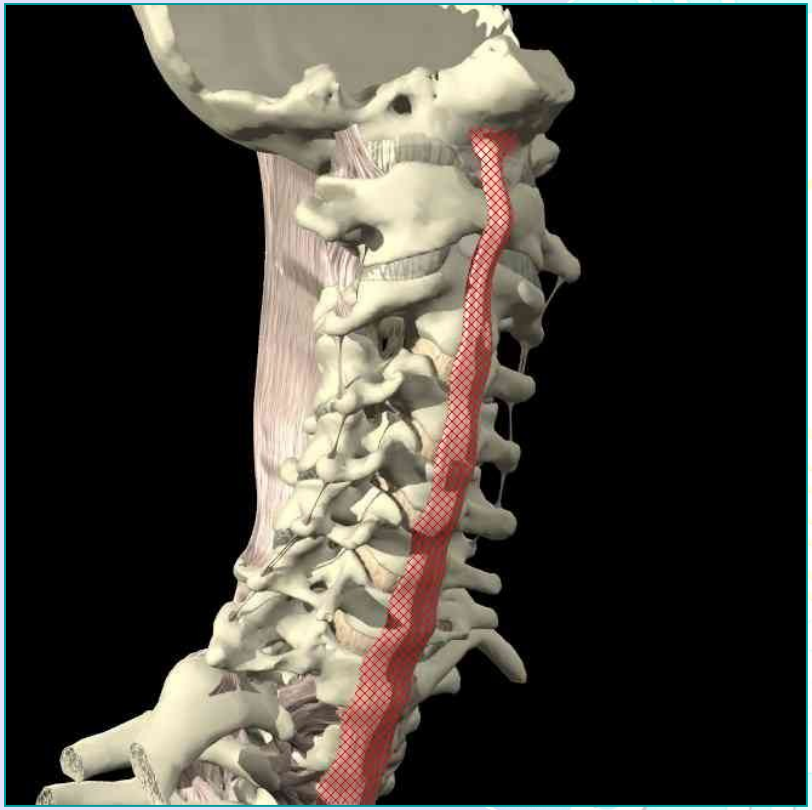
- Ligaments
  - The cervical spine also features a complex arrangement of ligaments to supplement its structure and mobility

LIGAMENTS



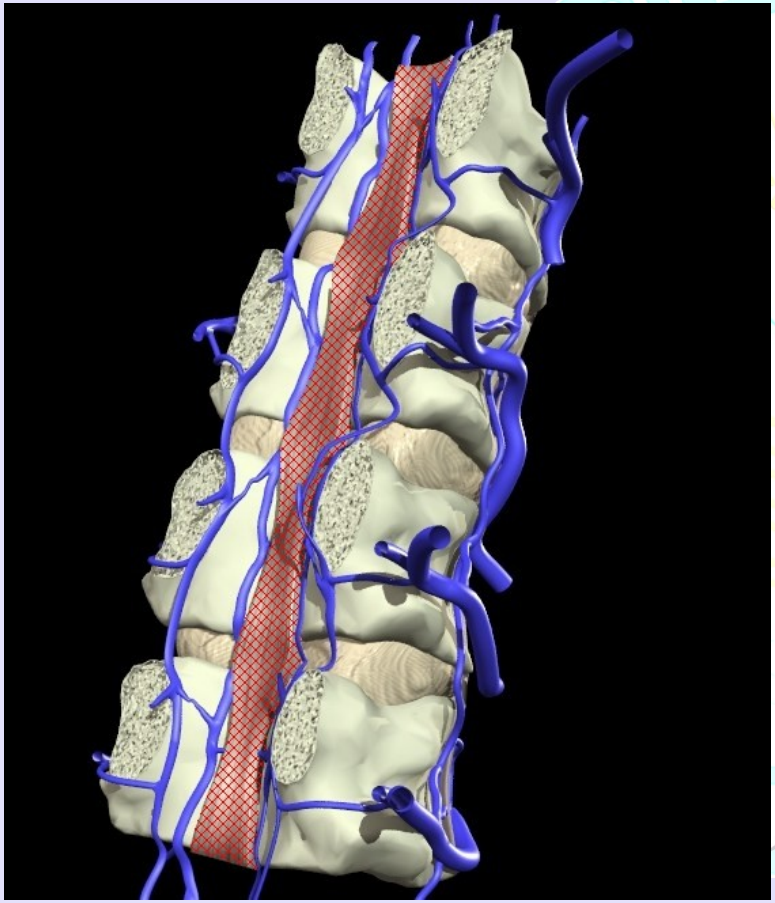
- Ligaments
  - ***Anterior longitudinal ligament***
  - Posterior longitudinal ligament
  - Ligamentum flavum
  - Intertransverse ligaments
  - Interspinous ligaments
  - Ligamentum nuchae

LIGAMENTS



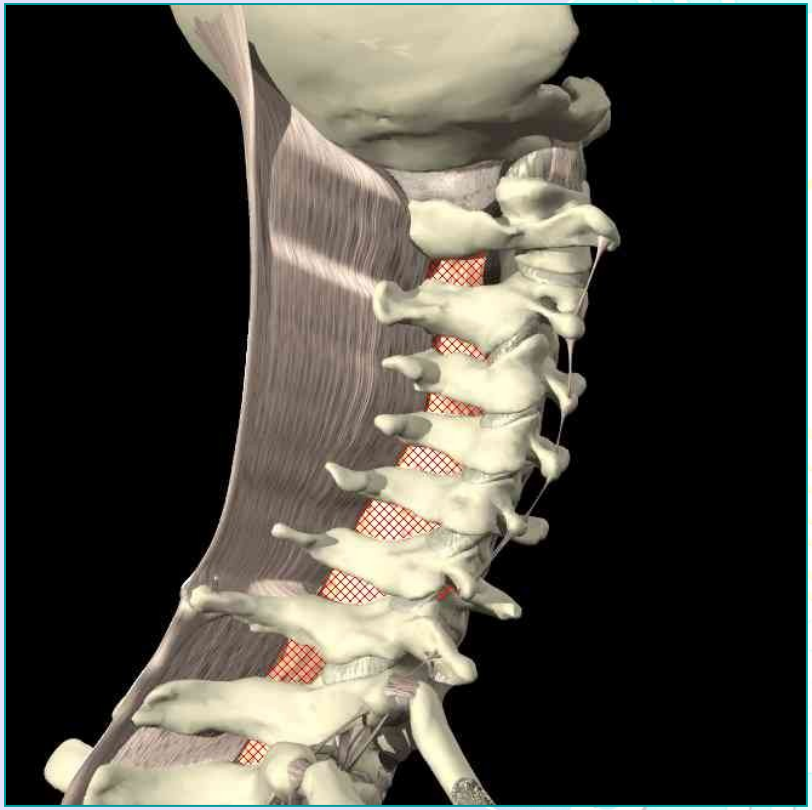
- Ligaments
  - ***Anterior longitudinal ligament***
  - Posterior longitudinal ligament
  - Ligamentum flavum
  - Intertransverse ligaments
  - Interspinous ligaments
  - Ligamentum nuchae

LIGAMENTS



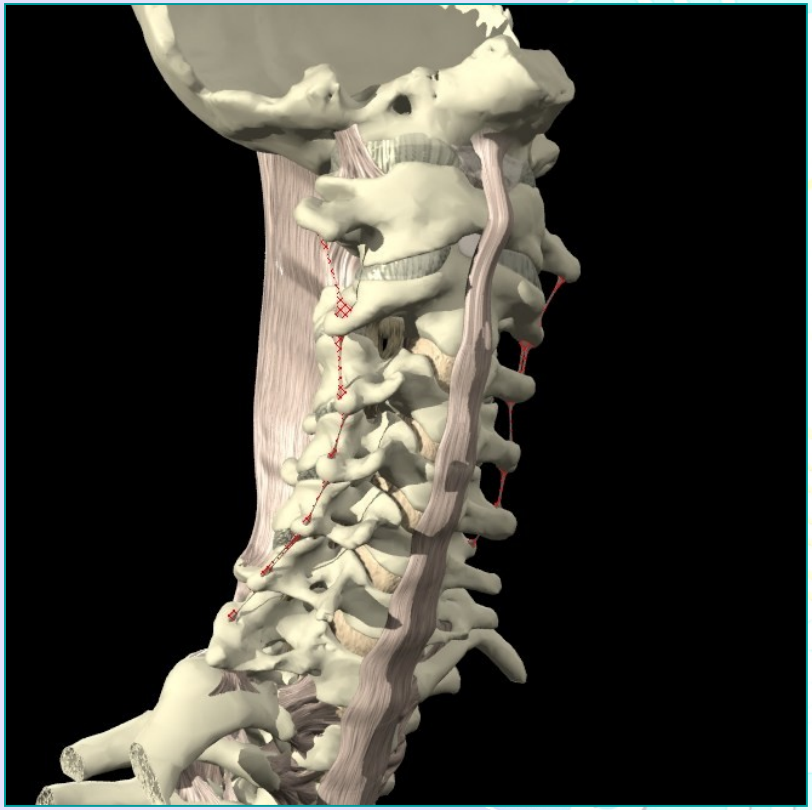
- Ligaments
  - Anterior longitudinal ligament
  - ***Posterior longitudinal ligament***
  - Ligamentum flavum
  - Intertransverse ligaments
  - Interspinous ligaments
  - Ligamentum nuchae

LIGAMENTS



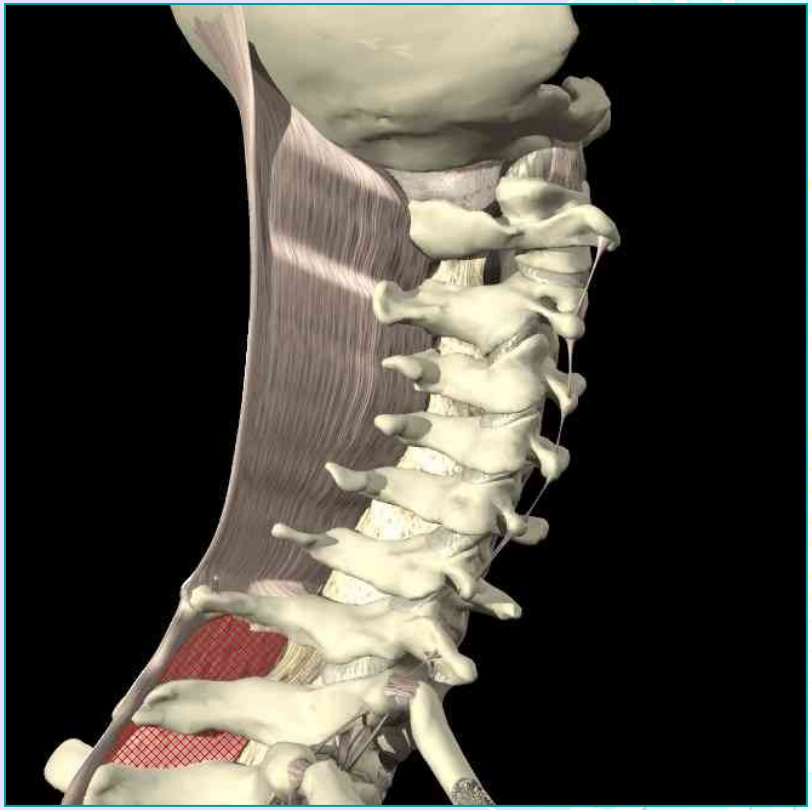
- Ligaments
  - Anterior longitudinal ligament
  - Posterior longitudinal ligament
  - ***Ligamentum flavum***
  - Intertransverse ligaments
  - Interspinous ligaments
  - Ligamentum nuchae

LIGAMENTS



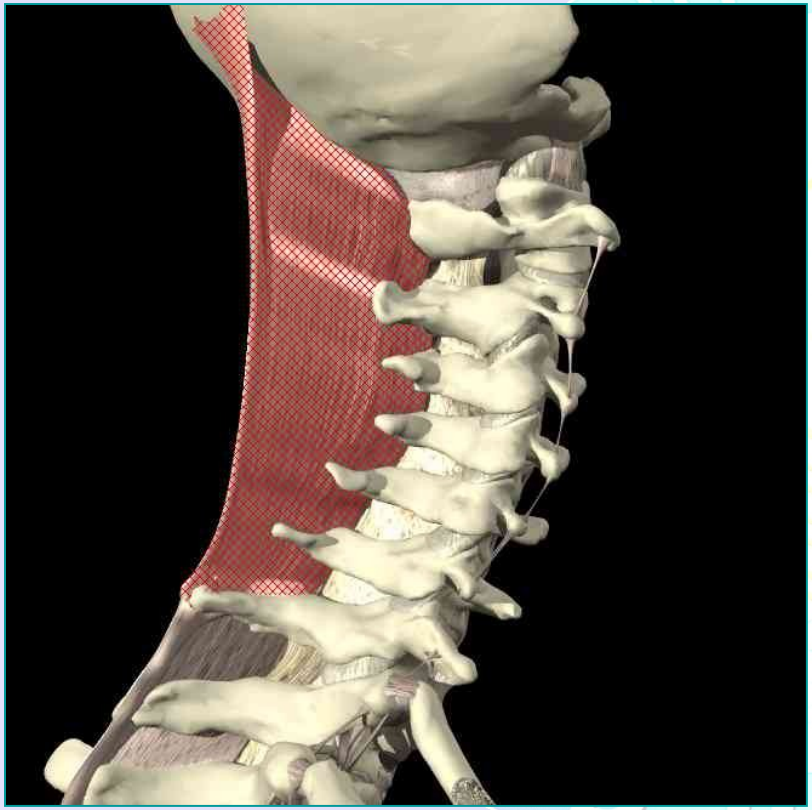
- Ligaments
  - Anterior longitudinal ligament
  - Posterior longitudinal ligament
  - Ligamentum flavum
  - ***Intertransverse ligaments***
  - Interspinous ligaments
  - Ligamentum nuchae

LIGAMENTS



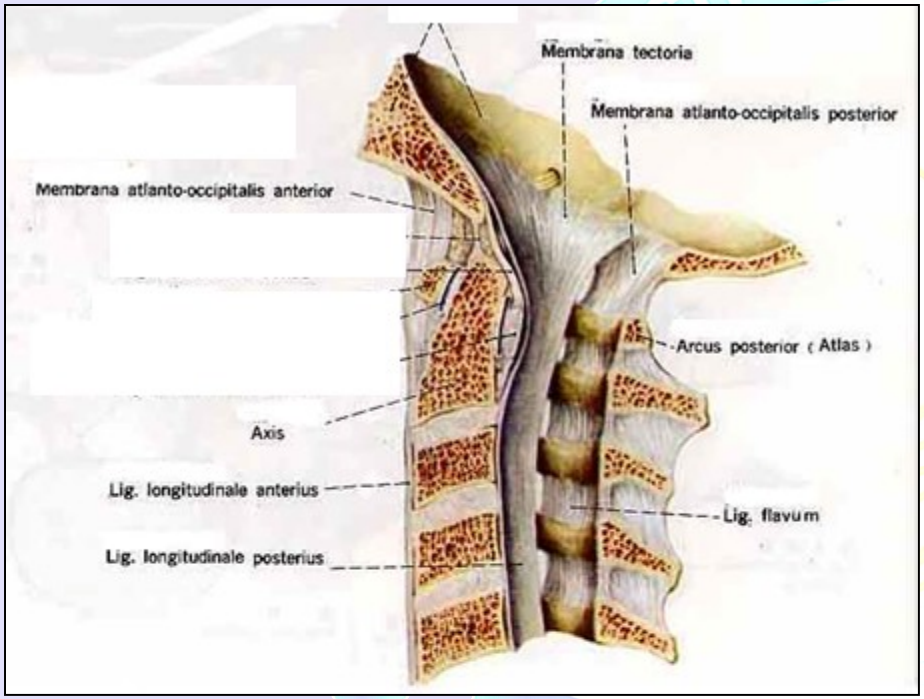
- Ligaments
  - Anterior longitudinal ligament
  - Posterior longitudinal ligament
  - Ligamentum flavum
  - Intertransverse ligaments
  - ***Interspinous ligaments***
  - Ligamentum nuchae

LIGAMENTS

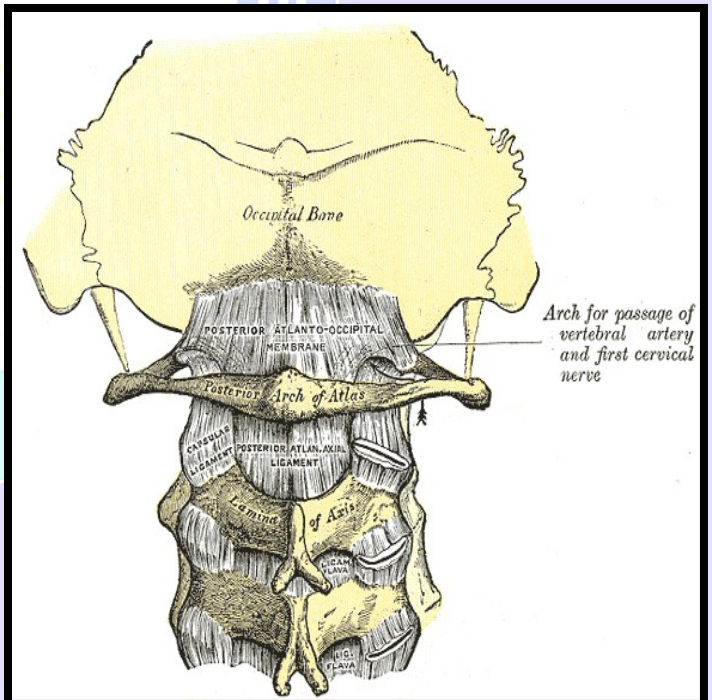


- Ligaments
  - Anterior longitudinal ligament
  - Posterior longitudinal ligament
  - Ligamentum flavum
  - Intertransverse ligaments
  - Interspinous ligaments
  - ***Ligamentum nuchae***

LIGAMENTS



- Anterior Atlanto-occipital membrane
- Posterior Atlanto-occipital membrane
- Membrena tectoria

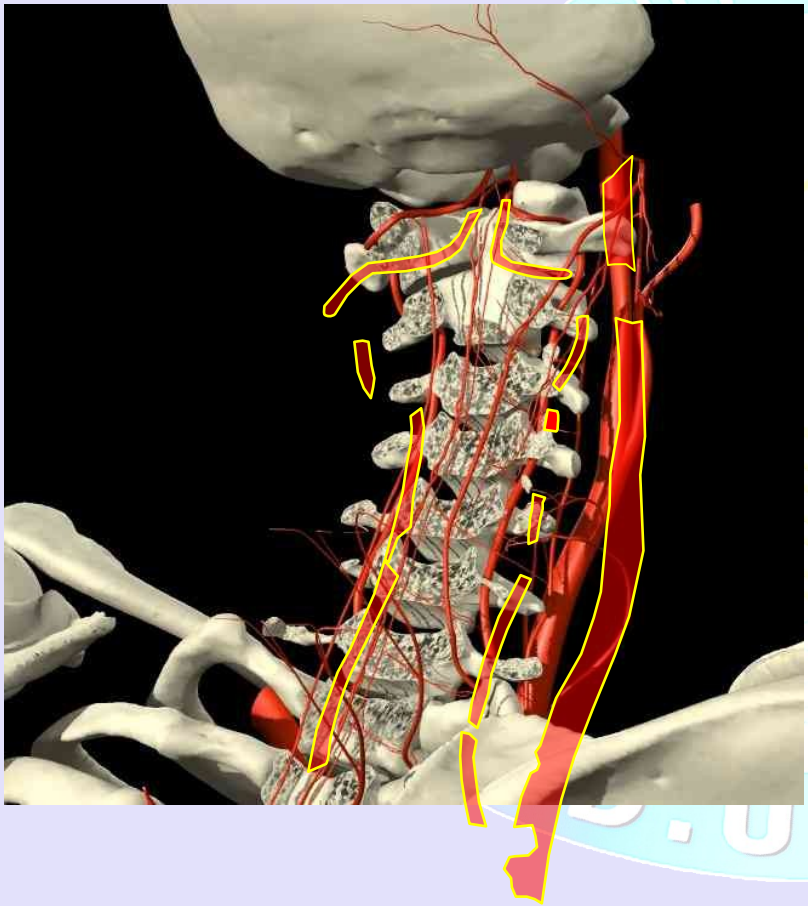


CERVICAL NERVES



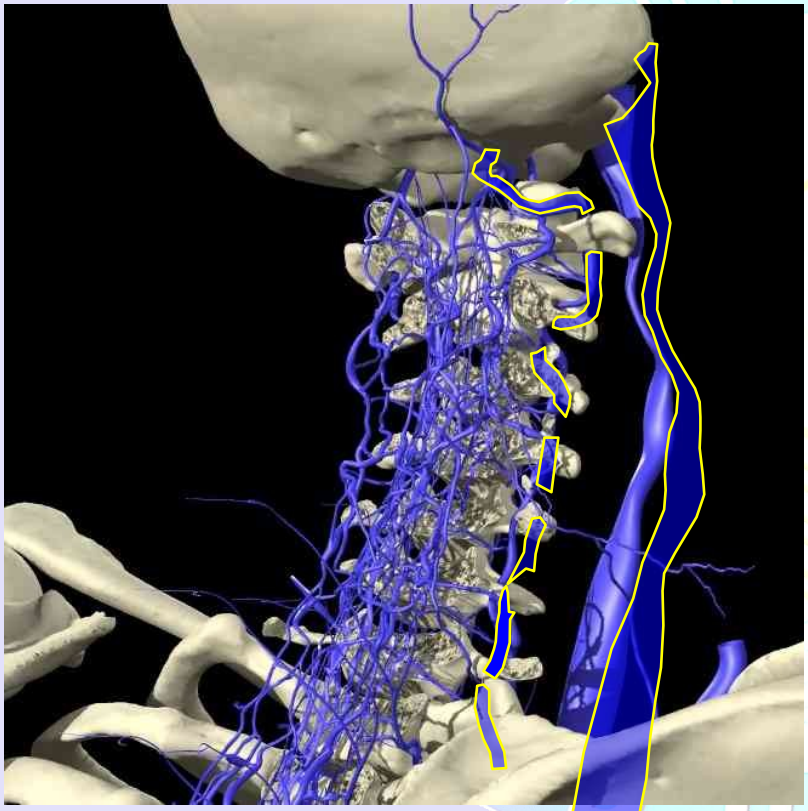
- Neural elements
  - 8 pair of cervical nerves
  - Exit the spinal canal superior to the vertebrae for which they are numbered
    - C1 nerves exit the canal between Occ & C1
    - C2 nerves exit the canal between C1 & C2
    - C8 nerves exit the canal between C7 & T1

RELATION OF MAJOR VESSELES



- Arteries
  - Carotid arteries
    - Located anterior and bilateral to the spine
  - Vertebral arteries
    - Enter the transverse foramen at C6 and continue through C1

RELATION OF MAJOR VESSELES

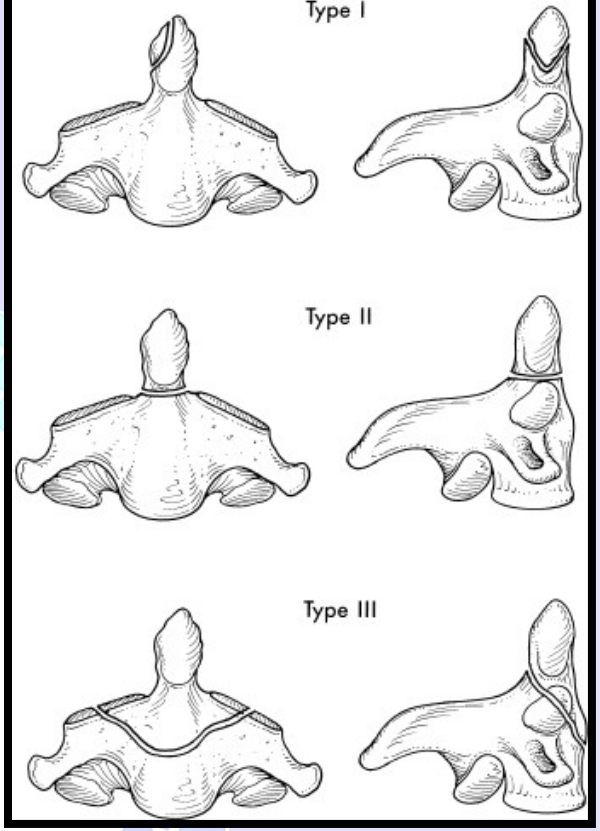


- Veins
  - Jugular veins
    - Located bilateral and anterior to the spine.
  - Vertebral veins
    - Located within the transverse foramen of C1-C7

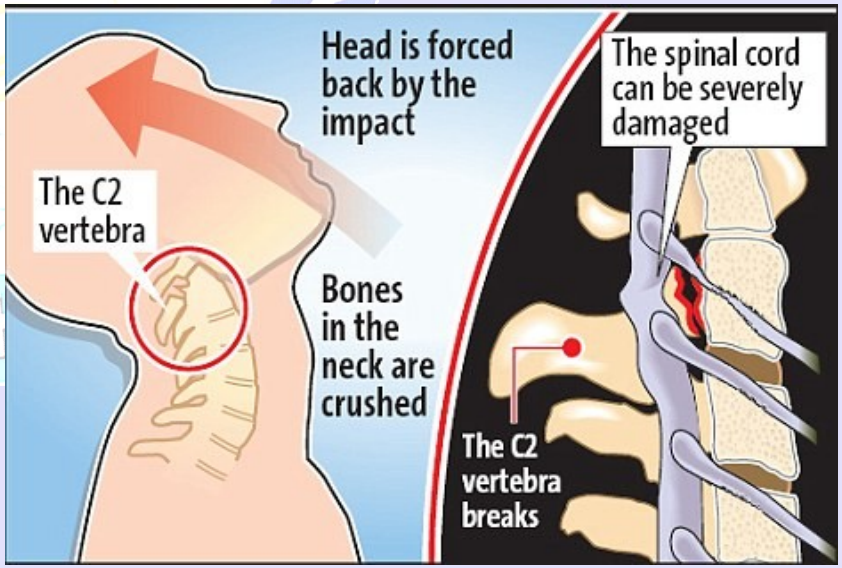
CLINICAL CO-RELATIONS

- Injuries are common at the level of the second cervical vertebrae.
- neurological injury is uncommon. If it does occur, however, it may cause death or profound disability, including paralysis of the arms, legs, and diaphragm, which leads to respiratory failure.

- **Common patterns of injury**
  - odontoid fracture
  - hangman's fracture fracture of both pedicles of the axis vertebra (C2).
  - treated with immobilization in a cervical collar or Halo brace.

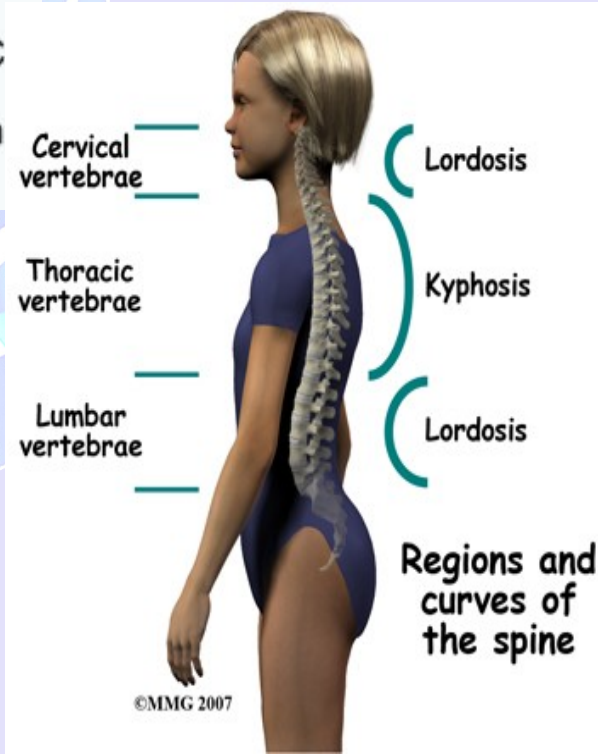




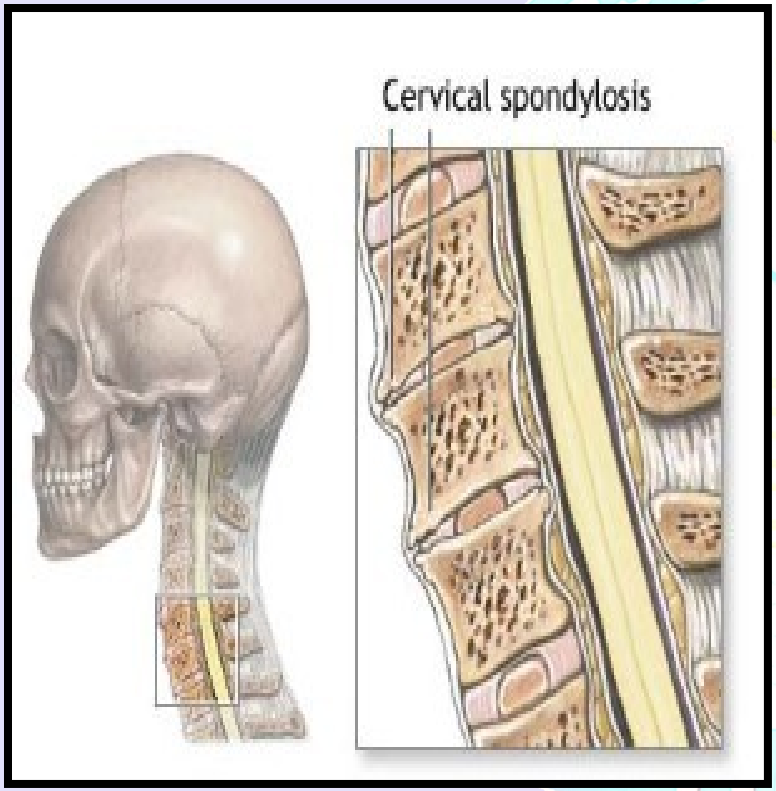


- ***Lordotic curve*** is an inward curvature of cervical vertebrae.
- Two segments of the vertebral column, namely cervical and lumbar, are *normally* lordotic, that is, they are set in a curve that has its convexity anteriorly and concavity posteriorly





CERVICAL SPONDYLOSIS



- Spondylosis is a disorder that is caused by the degeneration of the intervertebral discs.
- This results in the vertebrae rubbing together and exerting pressure on the nerves.
- **Cause:**
  - Chronic neck pain
- **Treatment:**
  - Conservative
  - surgical



THANK YOU